

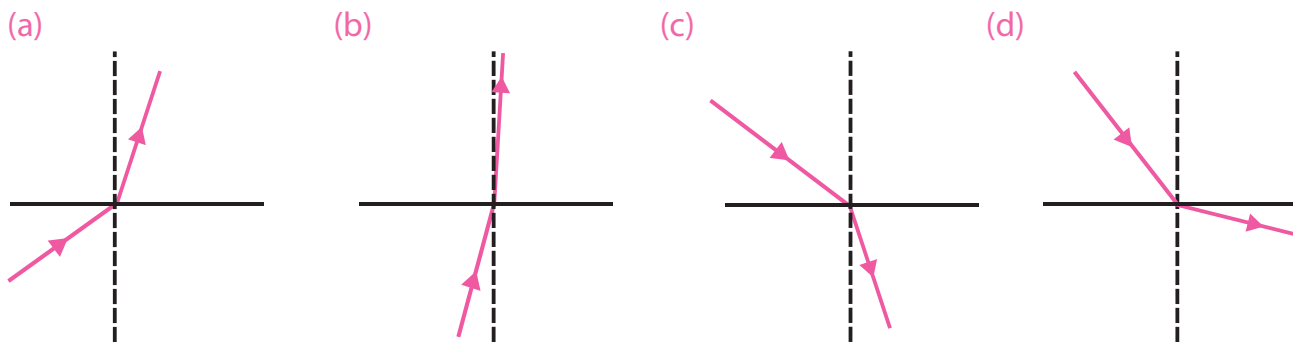
Refraction Practice

- 1 (a) When light crosses a surface from one transparent material to another, does it **always**, **sometimes**, or **never** refract? _____

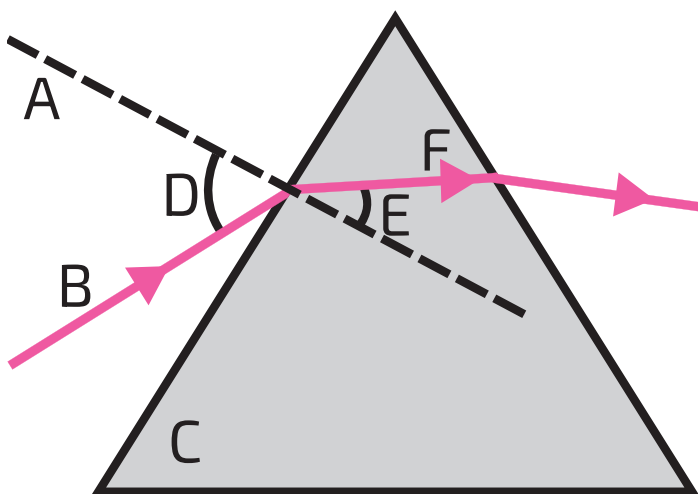
(b) Which factors decide **whether** refraction happens when light crosses a surface between materials?

- 2 A careless teacher has again forgotten to label their ray diagrams! One side of the boundary is air, one is glass.

Can you figure out which way around each of them goes? In the diagrams below, label each side of the surface.

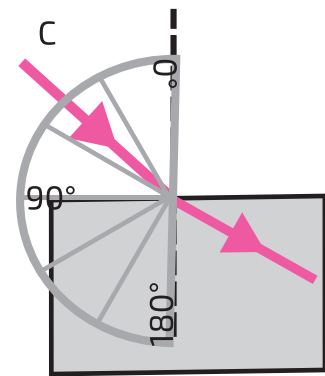
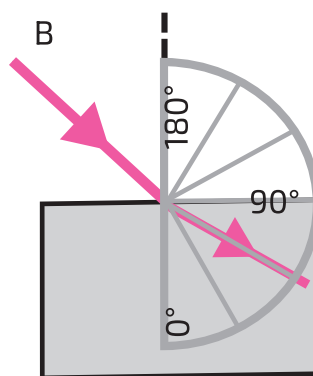
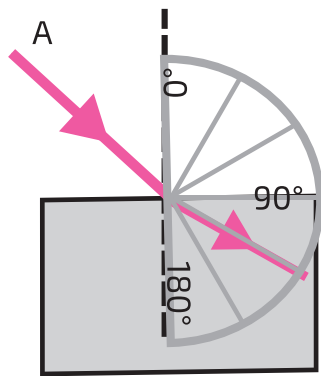


- 3 Which words describe the features in the diagram below? Fill in the table.



Technical term	Letter
glass block	
refracted ray	
incident ray	
angle of refraction	
angle of incidence	
normal	

4 The diagrams below show three attempts to measure an **angle of refraction**

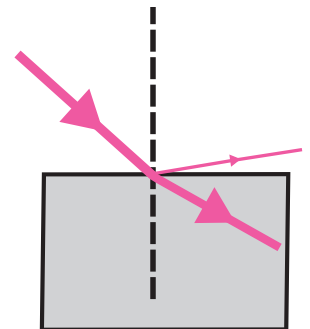


(a) From which of these protractors could we read the correct angle?

(b) State the angle of refraction of the ray.

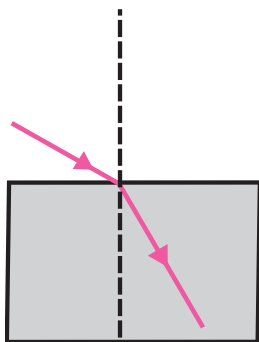
(c) Estimate the angle of incidence of the ray.

(d) Not all the light refracts. In the diagram on the right, someone has tried to draw the reflected ray. Spot and label the mistake.

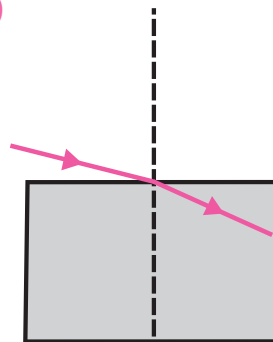


5 In each diagram below, write down the angles of incidence and refraction.

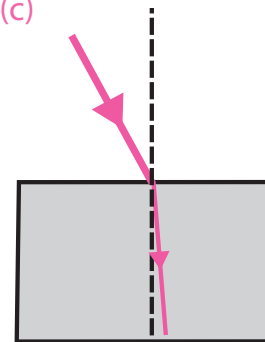
(a)



(b)

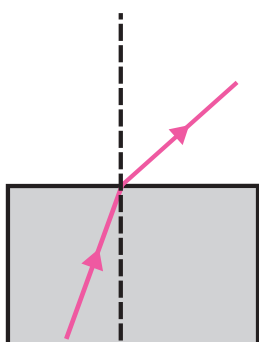


(c)

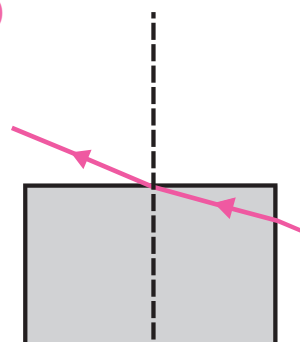


6 In each diagram below, write down the angles of incidence and refraction. Here the angles of incidence are inside the block of glass.

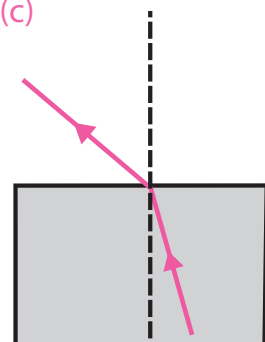
(a)



(b)



(c)



- 7 Complete the sentences using the phrases **fastest, slowest, towards, away from, speeds up, slows down, and change**. You do not need to use all of the phrases. It may help you to draw a diagram.

Light travels _____ in a vacuum and slows down when it enters glass.

If light hits glass at an angle, the side of the wave that hits first _____ first, causing the beam to _____ direction.

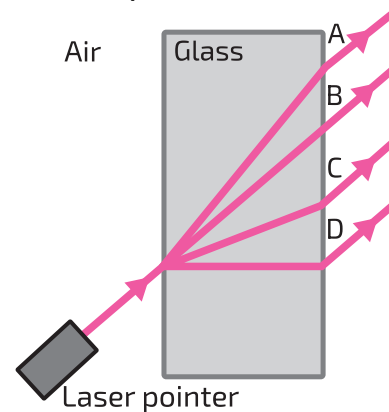
When light slows down, it bends _____ the normal.

When light speeds up as it exits the glass, it bends _____ the normal.

- 8 Some students are trying to predict the outcome of an experiment.

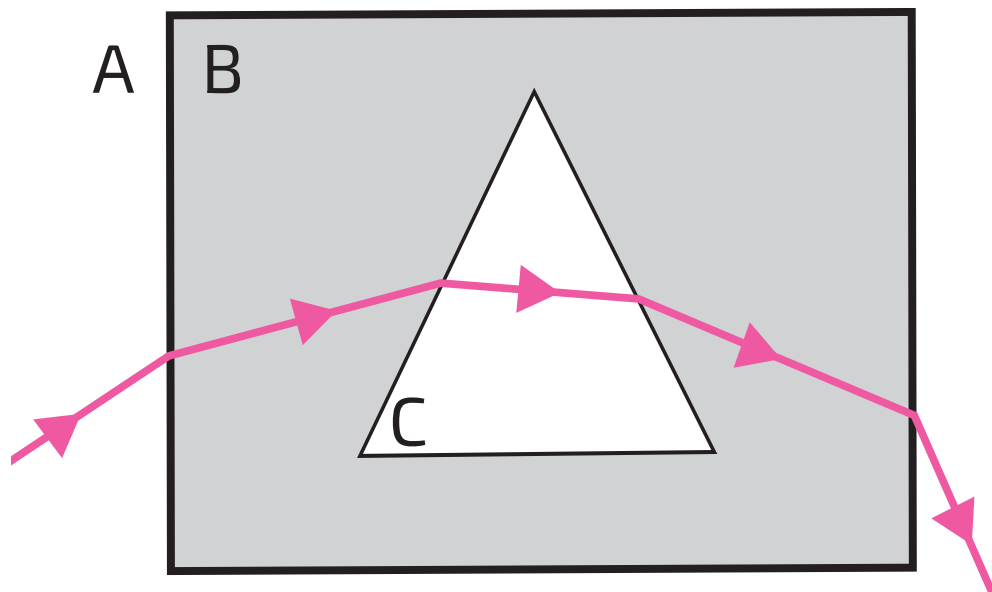
The diagram shows four different attempts at drawing what would happen when light shines through a glass block at an angle.

Which of the rays has been drawn correctly?



- 9 An antiques dealer has found a strange object in an old collection: a triangular prism sitting inside a rectangular prism.

To figure out how old it is, she wants to work out what type of materials were used to make it by studying how it interacts with light. The diagram shows a ray diagram for light passing through the object.



Order materials A, B and C according to the speed at which light travels through them, from slowest to fastest.